Unit VI

6 Domain Specific Applications of IoT

6.1 : Home Automation

Q.1 What do you mean home automation ? On which levels IoT works ?

Ans. : • Home automation is the automatic control of electronic devices in your home. These devices are connected to the Internet, which allows them to be controlled remotely.

- Interconnected devices enable to intelligently monitor and control smart homes in a future Internet of Things.
- Energy saving applications, for example, control indoor climate and electricity usage by employing context information to switch off appliances (e.g., lights, computers), reduce room temperature, close windows or stop warm water circulation.
- Home automation works on three levels :

1. **Monitoring :** Monitoring means that users can check in on their devices remotely through an app. For example, someone could view their live feed from a smart security camera.
2. **Control :** Control means that the user can control these devices remotely, like planning a security camera to see more of a living space.
3. **Automation :** Finally, automation means setting up devices to trigger one another, like having a smart siren go off whenever an armed security camera detects motion.

Q.2 Determine the IoT-levels for designing home automation IoT systems including smart lighting and intrusion detection ?

Ans. : • For designing home automation, IoT Level 1 is used.

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- The device Functional Group (FG) contains devices for monitoring and control. In the home automation example, the device FG includes a single board minicomputer, a light sensor and relay switch.
- The communication FG handles the communication for the IoT system. The communication FG includes the communication protocols that form the backbone of IoT systems and enable network connectivity. The communication API home automation example is a REST based APIs.
- By analyzing and sensing the human movements and environment, the light can be controlled by the smart lightening system. For example, a person enters a room, the light turns on automatically and it turns off when a person leaves the room.
- For this purpose, solid state lightening, IP enabled light are included. These can be controlled via mobile or web application. E.g. Phillips hue lights intrusion detection includes the sensors and cameras used to raise alerts and detect intrusions via SMS, image, video and email. This will improve security.

Q.3 How IoT helps for smart lighting ?

Ans. : • Smart control the lights with automation signal system to save energy. Smart, connected lighting is the next - generation energy - efficient LED products with additional sensors to sense things such as occupancy and temperature.

- In automatic light control system, Light Dependent Resistor (LDR) sensor is used to detect bright /medium /dim /dark conditions.
- It is simple enough to envision the addition of sensors and communications to create that initial concept of smarter, more adaptive lighting. If people are present, turn the lights on; if not, turn them off. Or use your smart phone to connect to the lighting system and tune it to the desired brightness level or to a particular color.
- Smart lighting is considered the one of the main solutions for energy reduction by means of controlling lighting level according to desired need with minimum energy consumption.

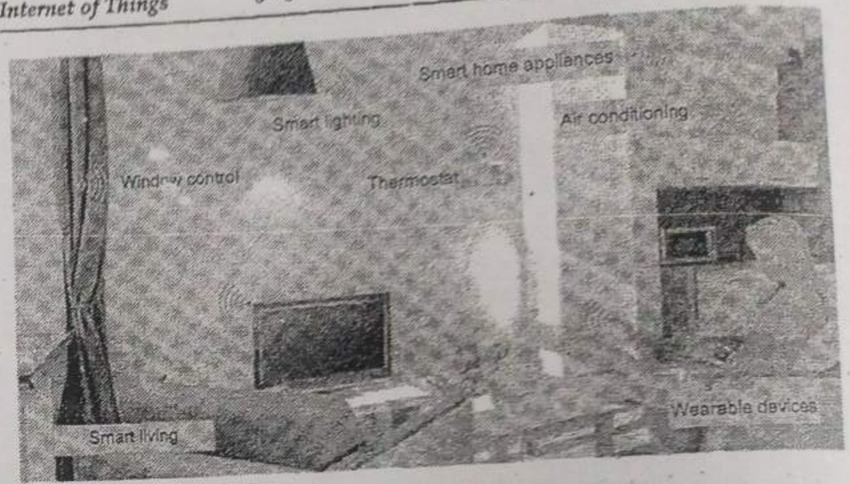


Fig. Q.3.1 Smart home

- Smart lighting systems utilize motion and light sensors for performing the control algorithms.
- The system uses motion and light sensors for detecting the surrounding environment. There are lamps controlled with the specific lighting level in order to supply the adequate amount of lighting required without affecting the user visibility.
- Certainly the required lighting level is strongly dependent on the weather conditions. In clear weather at night might require more luminance than cloudy one, due to the reflection from the clouds.
- While during mist and foggy weathers require the highest possible lighting level, as the visibility reaches its lowest. On snowy weather it might require an intermediate level between clear and foggy.
- During night it requires high lighting levels, while at day it needs just fade level to provide guidance or turn off if the weather is clear. The lighting concentration in the yard is affected by the above conditions.

Q.4 Explain working of intrusion detection using IoT ?

Ans. : • Intrusion Detection System (IDS) includes both hardware and software mechanisms and IDS is responsible for identifying malicious activities by monitoring network environment and system.

- The purpose of home intrusion detection system is to detect intrusions using sensors and raise alerts, if necessary.
- With the help of light dependent resistor and PIR motion sensor, it detect the motions in the room. If a motion is detected, system capture the image with the help of a webCam and store locally. Now the alerts are sent to the user with the captured image.
- Fig. Q.4.1 shows block diagram of intrusion detection.

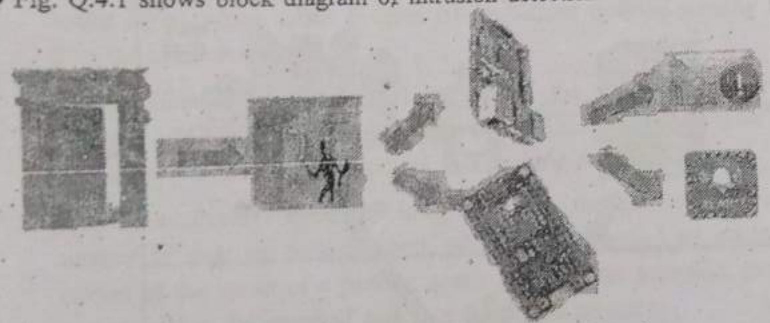


Fig. Q.4.1 Block diagram of intrusion detection

- To detect any form of intrusion in restricted areas and report it immediately, following concept is used.

1. A PIR sensor is required to detect the presence of any human being in the room.
2. An RFID is required to validate the presence of the person in the room by tallying his identity with those in the database.
3. A camera is required to click the picture of the room and send it via email as an alarm.
4. An internet connection is required to register all these movements on a website so that it can be accessed from any place and any device.

- The different input / output devices are controlled using TCP/IP over the IEEE 802.11 standard protocol. Data being gathered from sensors, such as PIR sensors, temperature sensors, IR transmitter and receiver is being processed on micro - controller as a server.
- Passive Infrared (PIR) Sensor : PIR sensor is an electronic sensing device that senses infrared (IR) light emitted from entities in its field of view and used to detect motion in its range. It is activated only in the security mode to detect any unwanted motion at the entrance. If any unwanted movement is detected then it will signal the microcontroller to take necessary steps.
- Alarm : It will only be activated in the security mode when some intruder is detected by the PIR motion sensor.
- Cloud controlled intrusion detection is possible by using location aware services. Here geo - location of each node is independently detected and stored in the cloud.
- Some intrusion detection system uses UPnP technology. It is based on image processing to recognize the intrusion.

6.2 : Smart City

Q.5 What is smart parking ? Explain process specification of smart parking ? How it is implemented at various levels ?

Ans. : • Traffic congestion is major problem in big cities. Searching for a parking space is a routine (and often frustrating) activity for many people in cities around the world.

- After finding parking space to the driver, he parks the vehicle, it maybe spend small amount of time to looking for a city council parking attendant to pay the parking fees.
- The smart parking system is designed by making use of some IoT supportable hardware's such as raspberry pi, auridino boards etc.
- Smart parking systems typically obtains information about available parking spaces in a particular geographic area and process is real - time to place vehicles at available positions.

- It involves using low-cost sensors, real-time data collection and mobile-phone-enabled automated payment systems that allow people to reserve parking in advance or very accurately predict where they will likely find a spot.
- When deployed as a system, smart parking thus reduces car emissions in urban centers by reducing the need for people to needlessly circle city blocks searching for parking.
- It also permits cities to carefully manage their parking supply smart parking helps one of the biggest problems on driving in urban areas; finding empty parking spaces and controlling illegal parking.
- Smart parking application can be accessed by drivers from smart phones, tables. Sensor is used for each parking slot, to detect whether the slot is empty or occupied.
- Local controller collect the information and send to server using Internet. Fig. Q.5.1 shows process specification for smart parking IoT system.

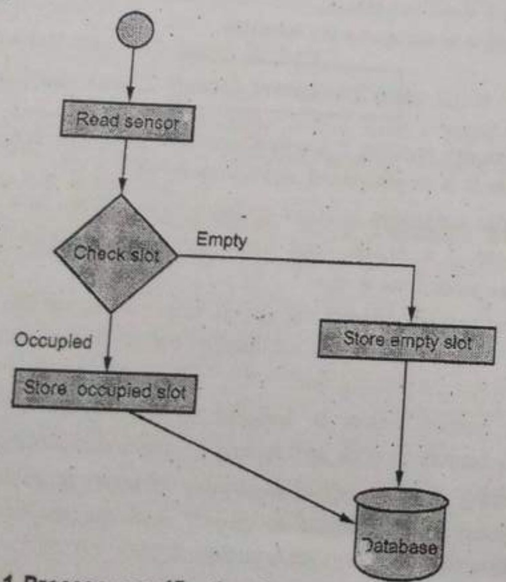


Fig. Q.5.1 Process specification for smart parking IoT system

- Each parking slot contains the sensor and it reads at regular intervals. Sensor sends the status information to local processing centre.
- Fig. Q.5.2 includes four layers : A sensing, networking, middleware and application layer.

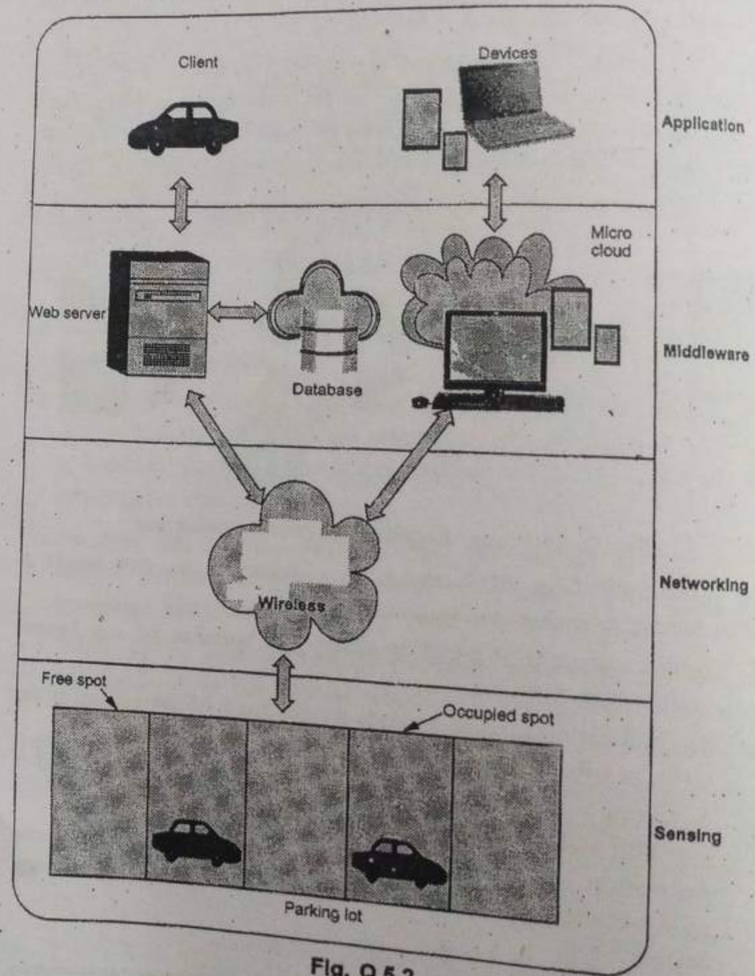


Fig. Q.5.2

- Sensing layer defines a platform where sensor devices are embedded into the parking lot to detect car presence/absence, and RFID devices located at the parking gates and strategic points of the parking are used to identify cars based on a unique mapping between RFID tags and car.
- Networking Layer : TCP/IP over Ethernet for connecting the gateway to the parking server and database and Internet access for remote access to the smart parking system from outside.
- Middleware layer hosts different databases and associated servers and manages all of the software intelligence provided by the smart parking system to provide smart services to users by enabling communication between the application layer where services are requested and the lower layers where smart devices are embedded into the parking lot to provide smart services.
- The application layer is the layer where the different services are defined and provided to different users. Client devices have been connected via the TCP/IP protocol to a parking database.
- Parking availability status by integrating into the car detection system sources of light on parking spots, which are controlled by actuators to inform of the status of a parking spot : E.g., red for occupied, green for empty, yellow for reserved and blue for out of service.
- Remote availability checking using the Internet and/or the GSM network to check in real time the availability of the smart parking system.
- The data of smart parking lots are able to provide profits for both customers and merchant's daily lives in the smart cities. This service works based on road sensors and intelligent displays which lead drivers to the best path for parking in the city.

Q.6 Write short note on smart road.

Ans. : • Sensor is installed on road to provides road traffic condition, travel time estimation, congestion and accident.

- Sensor collect this information and stored on the central database using cloud. This information helps for solving traffic congestion, making safe driving, keeping road condition upto date.

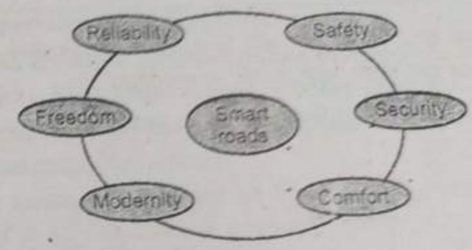


Fig. Q.6.1 Smart roads characteristics

- User can access the information from the cloud. User also get real time information.
- Real time traffic maps can be obtained to enable smooth flow. Traffic can be reduced with systems that detect alternate routes. User get timely information so they can locate a traffic free road, saving time and fuel. This information can reduce traffic jams and pollution improves the quality of life.

6.3 : Structural Health Monitoring

Q.7 Briefly discuss structural health monitoring.

Ans. : • The World Health Organization (WHO) defines E - health as : E - health is the transfer of health resources and health care by electronic means. It encompasses three main areas : The delivery of health information, for health professionals and health consumers, through the internet and telecommunications.

- E - health provides a new method for using health resources - such as information, money and medicines and in time should help to improve efficient use of these resources.
- E - health brings special characteristics. The monitoring device's environment is a patient; a living and breathing human being. This changes some of the dynamics of the situation. Human interaction with the device means batteries could be changed, problems could be filled in to technical support and possibly be resolved over the phone rather

than some type of service call. In most cases, the devices on the patient are mobile not static with regard to location.

- Fig. Q.7.1 shows high level E - health ecosystem architecture.

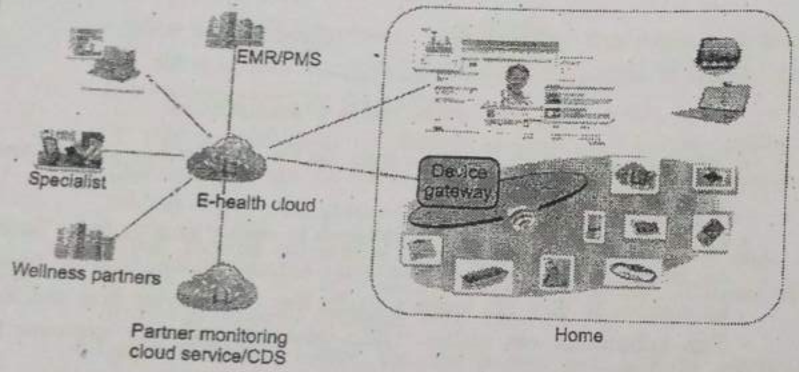
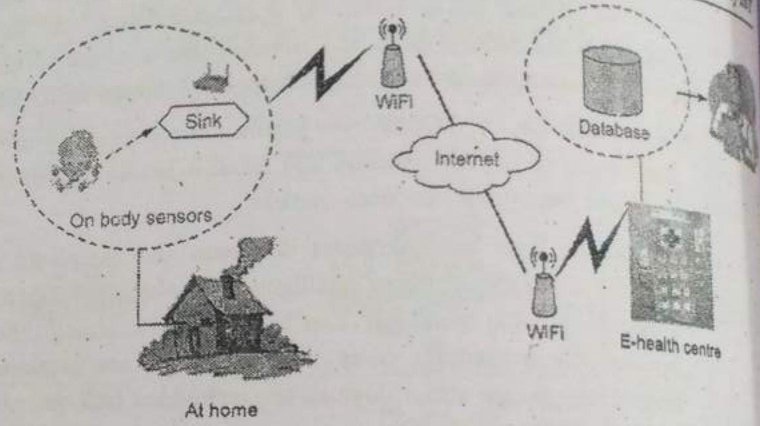


Fig. Q.7.1 High level E - health ecosystem architecture

- The data flow architecture focuses on the source of the data, the destination the data and path the data. The source of the data is typically the sensor.
- The data can be either locally cached or is sent to the upstream systems without storing in the sensor. The path taken by the data includes a gateway, which can also cache some of the data and do distributed processing.
- Intermediate hubs can also store and process the data to filter out or make certain decisions. A distributed rules engine is used to make distributed decisions at the closest point of care. This enables data traffic to be filtered and processed efficiently without having every data being processed by the cloud service.



- The development of wireless networks has led to the emergency of a new type of E - healthcare system, providing expert-based medical treatment remotely on time.
- With the E - healthcare system, wearable sensors and portable wireless devices can automatically monitor individuals' health status and forward them to the hospitals, doctors and related people.
- The system offers great conveniences to both patients and health care providers. For the patients, the foremost advantage is to reduce the waiting time of diagnosis and medical treatment, since they can deliver the emergent accident information to their doctors even if they are far away from the hospital or they don't notice their health condition.
- In addition, E - health system causes little interruption to patient's daily activities. For the health care providers, after receiving the abnormal signals from the patients, appropriate treatment can be made, which saves medical resources.
- Furthermore, without direct contact with medical facilities, medical personnel or other patients, the patients are unlikely to be infected with other diseases.

- However, to ensure the security and privacy of patient's medical records encounters a lot of challenges :

1. How to achieve the confidentiality and integrity of patients' information.
2. The security of wireless body area network.
3. The privacy and unlink ability of patients' health status.
4. The undeniability and unlinkability of doctors' treatment.
5. The location privacy of patients, the fine - grained access control of patient's medical record, the mutual authentication between patients and hospitals, etc.

- It would be useful to create an up-to-date bibliography on secure E - healthcare systems.

6.4 : Environment

Q.8 Explain weather monitoring system with the help of IoT.

Ans. : • Remote weather monitoring system which measures the following weather parameters :

1. Daylight : Using a photodiode as a wired binary switch sensor.
2. Humidity : Using a wired analog humidity sensor.
3. Temperature : Using a digital bit-stream temperature sensor via wireless medium employing wireless RF modules.

- The sensors are the miniaturized electronic devices used to measure the physical and environmental parameters. By using the sensors for monitoring the weather conditions, the results will be accurate and the entire system will be faster and less power consuming.
- The system monitors the weather conditions and updates the information to the web page. The reason behind sending the data to the web page is to maintain the weather conditions of a particular place can be known anywhere in the world. The weather condition is also displayed on the systems LCD.

Q.9 Explain air pollution monitoring system.

Ans. : • The rapid growth in infrastructure and industrial plants creating environmental issues like climate change, malfunctioning and pollution has greatly influenced for the need of an efficient, cheap, operationally adaptable and smart monitoring systems.

- Air monitoring sensor placed throughout the city. Sensor send periodic measurement of air quality data to a gateway.

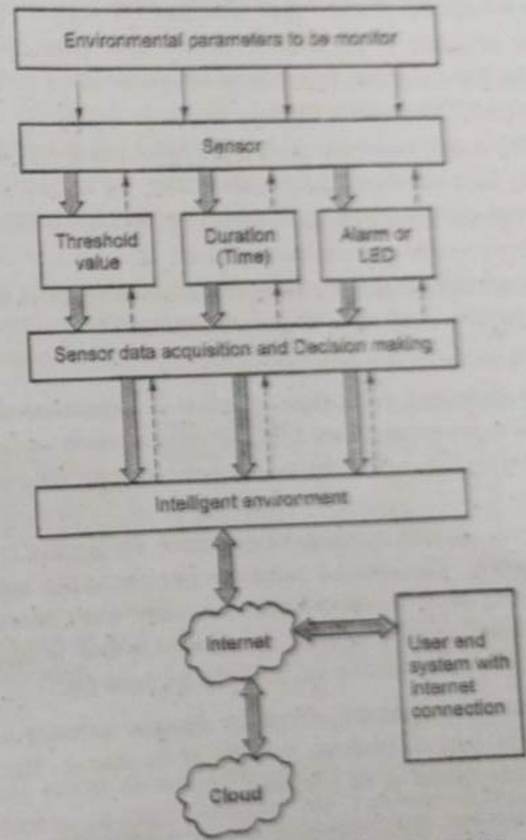


Fig. Q.9.1 Air pollution monitoring system

- Gateway sends information to network where the data is analyzed by the application server which can identify zones of concern and provide recommendations.
- Application server provides information regarding air quality level throughout the city, including alert and pollution pattern via computer or mobile device.
- Fig. Q.9.1 shows air pollution monitoring system.
- " The MQ series of gas sensors utilizes a small heater inside with an electro chemical sensor these sensors are sensitive to a range of gasses are used at room temperature.
- MQ135 alcohol sensor is a Sno2 with a lower conductivity of clean air. When the target explosive gas exists, then the sensor's conductivity increases more increasing more along with the gas concentration rising levels.
- By using simple electronic circuits, it convert the charge of conductivity to correspond output signal of gas concentration.

Q.10 Discuss forest fire detection system.

Ans. : • Forest fires cost millions of dollars in damages and claim many human lives every year. Apart from preventive measures, early detection and suppression of fires is the only way to minimize the damages and casualties.

- The forecast forest fires cannot be detected by the satellites fire spreads uncontrollable. The wireless sensor network can detect and forecast forest fire more perfectly and accurately than the traditional satellite-based detection approach. This method is used to minimize the loss of forests, wild animals and people in the forest fire.
- The most critical issue in a forest fire detection system is immediate response in order to minimize the scale of the disaster. This requires constant surveillance of the forest area.
- Current medium and large-scale fire surveillance systems do not accomplish timely detection due to low resolution and long period of scan. Therefore, there is a need for a scalable solution that can provide

real time fire detection with high accuracy. We believe that Wireless Sensor Networks (WSN) can potentially provide such solution.

- Network is composed of numerous and ubiquitous micro sensor nodes which have the ability to communicate and calculate. These nodes can monitor sense and collect information of different environments and various monitoring objects co-operatively.
- A large number of sensor nodes are deployed in a forest to detect fire. Those are mainly cluster heads, sink, wind sensor and managing node.
- A few wind sensor nodes are manually connected to the sink via wired networks to detect wind speed. Sensor nodes collect measured data for example, environment temperature, relative humidity and smoke.
- A sensor node is commonly composed of a sensor module, a processing module, a wireless communication module and a power module. The temperature sensor used is thermostat.
- The main need for thermostat is to detect forest fire. The advantage of using thermostat is that it helps in knowing the temperature variations in several areas of the forest, whereas the fire sensor will detect only fire and will not give the temperature variations of those areas.

6.5 : Logistics

Q.11 Explain fleet tracking ? What are the advantages of fleet tracking system ?

Ans. : • It is automated vehicle routing and scheduling. It supports driver compliance, safety and performance reporting.

- Vehicle fleet tracking system uses GPS technology to track the location of the vehicle in the real time.
- Fleet maintenance and fuel conservation capabilities. Track, schedule and route vehicles in real time.
- Proactively manage fleet maintenance and fuel economy. It is also possible to monitor driver behavior and performance (distance traveled, speed, location).

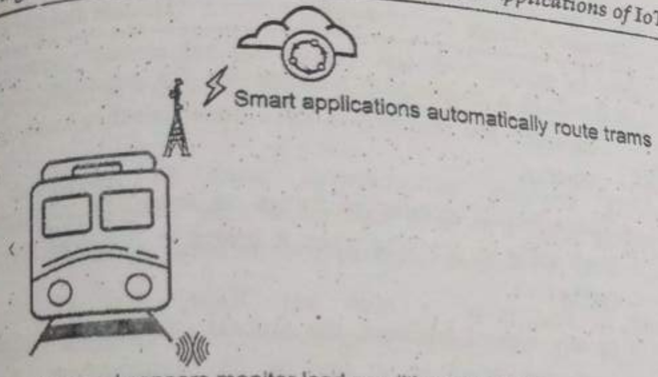


Fig. Q.11.1

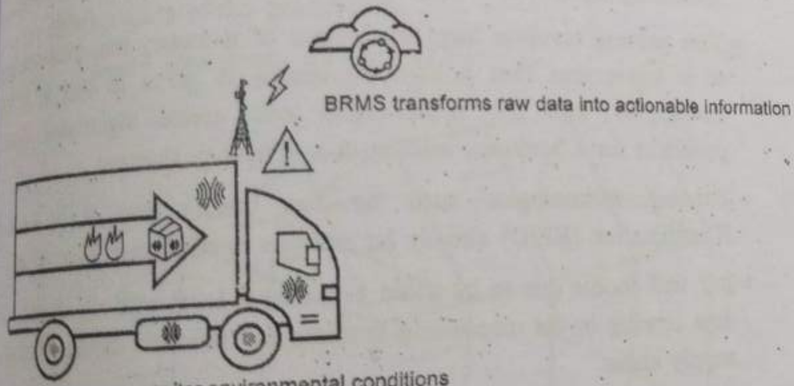


Fig. Q.11.2

Benefits :

a) Accelerate delivery and dispatch rates.

b) Improve customer satisfaction.

c) Reduce fuel consumption and vehicle maintenance costs.

d) Ensure compliance with government and industry regulations.

e) Improve fleet productivity, uptime and safety.

Q.12 How IoT helps for remote vehicle diagnostics.

Ans. : • Remote vehicle diagnostics solution monitors the health of the vehicle, determines the root cause of the problem / failure and provides

real time information of vehicle parameters to assess its performance against benchmarks.

- The solution monitors the health of the electric vehicle, commercial vehicle, utility vehicle and provides insight to field support staff to determine the root cause of the problem. It also enables the customers to access information about the vehicle. Commercial / Utility vehicles being driven across the country extensively over time for various purposes are in need of a diagnostic check which is automated through the offering.
- By monitoring all the aspects of the car is easier to detect any problem in advance by sending all, sensor readings to a certified center where technicians and engineers will apply their expertise to find and predict imminent failures of key systems integrated in the vehicle.
- Modern commercial vehicles support on board diagnostic standard. Next generation vehicles will have sophisticated on-board connectivity equipment, providing wireless network access to the vehicle for infotainment and other telematics services. Fig Q.12.1 shows remote vehicle diagnostics.

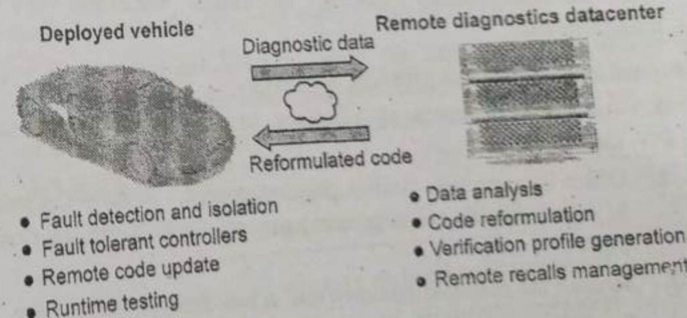


Fig. Q.12.1 Remote vehicle diagnostics solution

- In vehicle, sensors connect to the vehicle terminal which is responsible for collecting, storing, processing and reporting information and responding to commands from supervision platforms.

- The vehicle terminal consists of the microprocessor, data storage, GPS module, wireless communication transmission module, real time clock and data communication interface.

6.6 : Retails

Q.13 Write short note on smart vending machine.

Ans. : • Smart vending is about building remote management systems and telemetry tools, which integrate monitoring, transmission and delivery of operational data from each vending machine via the Internet.

- Smart vending solution offers its customer's flexible payment options and monitors the machines remotely and in real time.
- Smart phone applications that communicate with smart vending machine allow user preferences to be remembered and learned with time.
- For instance, Innovations like RFID based "smart" shelves continuously scan items on the shelf and notifies the appropriate systems. During low or out of stock situations they create automatic replenishment alerts and send automatic orders directly to central warehouse and to manufacturers.
- Smart vending machine provided following :
  1. Achieve high levels of efficiency in the management of their assets.
  2. Offers its customer's flexible payment options : RFID/NFC Card; - Mobile Payments; - Smartphone payments; Cash; Debit and Credit Card.
  3. Monitor the machines remotely and in real time.
  4. Simplifies business since the vending machines contain multiple sensors that alert the owners about their location, the state inventory and eventual maintenance issues.

Q.14 Discuss inventory management system using IoT.

Ans. : • Retail involves the sale of goods from a single point (malls, markets, department stores etc) directly to the consumer in small quantities for his end use.

- Retail is a challenging business but the pressures of today's economic conditions are resulting in even more selective consumer shopping and spending.
- The effect of internet of things on inventory management is the next huge thing in progress when it comes to Business Process Management (BPM).
- In any typical business, the process of ordering, storing, tracking and managing good is a day to day requirement. As with all high investment top-tier businesses, this process becomes more complex with increasing amount of supply and demand.
- This process involves huge transaction of monetary resources and hence it is impervious that a high preference is given to this in a BPM. Inventories that are mismanaged can create significant financial problems for a business, leading to a inventory shortage.
- Existing technologies such as bar coding and Radio-Frequency IDentification (RFID) already let retailers monitor their inventories.
- IoT will enable this to be taken to the next level with significantly more data coming in the monitoring systems and products moving through the supply chain.
- This can considerably improve supply chain efficiencies and enable leaner inventories. Large retailers such as Walmart are already using IoT for supply chain and inventory management.
- Tracking is done using RFID readers attached to the retail store shelves.

6.7 : Agriculture

Q.15 Discuss controlling greenhouse using IoT.

Ans. : • In modern greenhouses, several measurement points are required to trace down the local climate parameters in different parts of the big greenhouse to make the greenhouse automation system work properly.

- The most important factors for the quality and productivity of plant growth are temperature, humidity, light and the level of the carbon dioxide.
- Continuous monitoring of these environmental variables gives information to the grower to better understand, how each factor affects growth and how to manage maximal crop productiveness.
- Wireless Sensor Network (WSN) can form a useful part of the automation system architecture in modern greenhouses.
- Wireless communication can be used to collect the measurements and to communicate between the centralized control and the actuators located to the different parts of the greenhouse.
- Fig. Q.15.1 shows greenhouse with sensor.

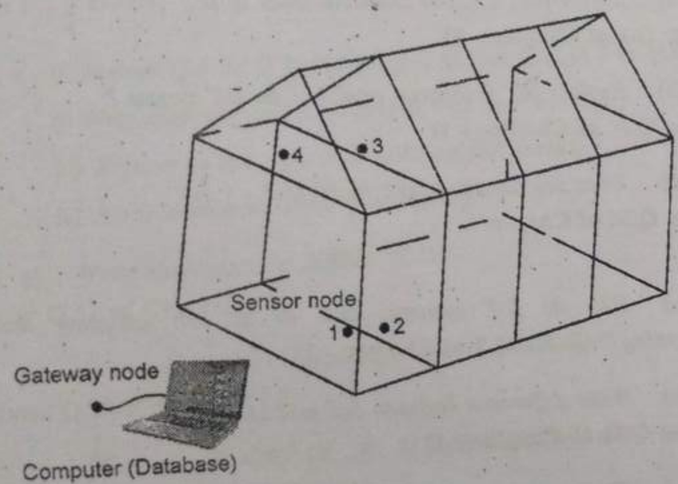


Fig. Q.15.1 Greenhouse with sensor

- Basic factors affecting plant growth are sunlight, water content in soil, temperature, CO_2 concentration etc. These physical factors are hard to control manually inside a greenhouse and there is a need for automated design arises.

- Data collected from various sensor is stored on the centralized server and this server process the data.

Q.16 Write short note on smart irrigation.

Ans. : • In our country, agriculture is major source of food production to the growing demand of human population. In agriculture, irrigation is an essential process that influences crop production.

- Generally farmers visit their agriculture fields periodically to check soil moisture level and based on requirement water is pumped by motors to irrigate respective fields.
- The smart irrigation system was developed to optimize water use for agricultural crops. The system has a distributed wireless network of soil-moisture and temperature sensors placed in the root zone of the plants.
- Wireless Transmitter Unit (WTU) is comprised of a soil moisture sensor, a temperature sensor, a microcontroller, a RF transceiver and power source. Several WTUs can be incorporated in field to form a distributed network of sensors.
- Input to the micro controller is the reading of the moisture sensor and depending upon the threshold value a high or a low.
- If the soil moisture value is below the threshold or the temperature exceeds the threshold value, then the motor is turned on till the levels of moisture and temperature are optimized. Otherwise the motor is off. The sensor values and motor status is displayed on an Android App.

6.8 : Industry Applications

Q.17 Write short note on machine diagnosis and prognosis.

Ans. : • Machine fault diagnostic and prognostic techniques have been the considerable subjects of condition-based maintenance system in the recent time due to the potential advantages that could be gained from reducing downtime, decreasing maintenance costs, and increasing machine availability.

- A failure in industrial equipment results in not only the loss of productivity but also timely services to customers and may even lead to safety and environmental problems.
- IoT play an important role in both diagnosis and prognosis. Critical manufacturing processes and equipment must be continuously monitoring for any variations or malfunctions. A slight shift in performance can affect overall product quality or manufacturing equipment health.
- With group of sensing nodes monitoring various manufacturing equipments and processes and transmitting data in periodic manner, situations may arise where the engineer might want to query data from some specific nodes to estimate current status of particular process or equipment.
- There can be situation of unforeseen malfunctioning or variations beyond prescribed tolerance bands. A mechanism is hence required to define tolerance bands for each sensing module. When measurements at particular node exceed the tolerance, the node must breach the periodic cycle to send an alarm about the emergency.
- Case Based Reasoning (CBR) is normally used method to find solution to new problems based on previous experience.
- CBR is an effective method for problem solving for quantitative mathematical model i.e. machine diagnostic and prognosis.

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